Algebra 8 Placement Test — ANSWER KEY

Sienna Ash · 100 points total

Grader use only. Worked solutions shown for every item. Bold red = expected answer.

Section 1 — Evaluating Expressions & Function Notation

14 pts

Q1.

2 pts

Evaluate $3x - 5$ when $x = 4$.

$3(4) - 5 = 12 - 5 = \mathbf{7}$

Q2.

4 pts

Evaluate $2a^2 - 3b$ when $a = -2$ and $b = 5$.

$2(-2)^2 - 3(5) = 2(4) - 15 = 8 - 15 = \mathbf{-7}$

Q3.

4 pts

Evaluate $\dfrac{5x - y}{2}$ when $x = 3$ and $y = -1$.

$\dfrac{5(3) - (-1)}{2} = \dfrac{15 + 1}{2} = \dfrac{16}{2} = \mathbf{8}$

Q4.

4 pts

Let $f(x) = 3x - 1$.

(a) Find $f(-2)$.

$f(-2) = 3(-2) - 1 = -6 - 1 = \mathbf{-7}$

(b) Find the value of x where $f(x) = 11$.

$3x - 1 = 11 \;\Rightarrow\; 3x = 12 \;\Rightarrow\; x = \mathbf{4}$

Section 2 — Solving Equations

30 pts

Q5.

4 pts

$2x + 9 = 21$

$2x = 12 \;\Rightarrow\; x = \mathbf{6}$

Q6.

4 pts

$3(x-4)=15$

$3x-12=15 \;\Rightarrow\; 3x=27 \;\Rightarrow\; x=\mathbf{9}$

Q7. 6 pts

$4x-7=x+8$

$4x-x=8+7 \;\Rightarrow\; 3x=15 \;\Rightarrow\; x=\mathbf{5}$

Q8. 6 pts

$2(3x+1)=4x-6$

$6x+2=4x-6 \;\Rightarrow\; 2x=-8 \;\Rightarrow\; x=\mathbf{-4}$

Q9. 4 pts

$\frac{2}{3}x-4=6$

$\frac{2}{3}x=10 \;\Rightarrow\; x=10\cdot\frac{3}{2} \;\Rightarrow\; x=\mathbf{15}$

Q10. 3 pts

$5x+2=5x-3$

Subtract $5x$: $2=-3$ — false. **No solution.**

Q11. 3 pts

$4(x-1)=4x-4$

Distribute: $4x-4=4x-4 \;\Rightarrow\; 0=0$ — always true. **Infinitely many solutions.**

Section 3 — Proportional Relationships

18 pts

Q12. 6 pts

Table: (0,3), (2,7), (4,11), (6,15)

(a) **Not proportional** — when $x=0$, $y=3\neq 0$. A proportional relationship passes through the origin.
(b) y-intercept = **3**
(c) Slope: $\Delta y/\Delta x=4/2=2$. Equation: $y=2x+3$

Q13. 6 pts

Line through $(0,-2)$ and $(4,6)$.

(a) **Not proportional** — y-intercept is -2, not 0.

(b) $m = \dfrac{6-(-2)}{4-0} = \dfrac{8}{4} = \mathbf{2}$

(c) $y = 2x - 2$

Q14. 6 pts

Parking garage: \ entry + \/hour.

(a) $C = 3h + 5$

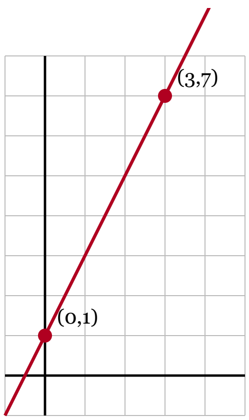
(b) Slope = **cost increases by \ per additional hour** (the per-hour rate).

(c) **Not proportional** — the \ flat fee means $C = 5$ when $h = 0$, so the y-intercept is 5, not 0.

Section 4 — Identifying Slope 20 pts

Q15. 4 pts

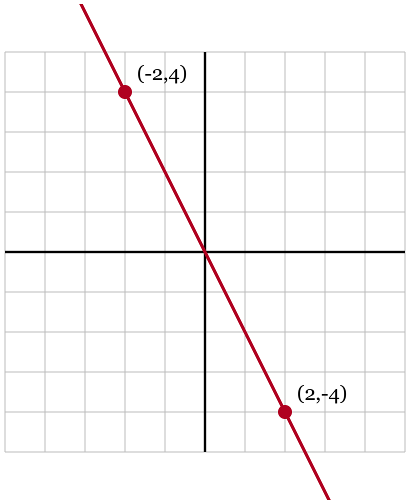
Line through $(0, 1)$ and $(3, 7)$.



$m = \dfrac{7-1}{3-0} = \dfrac{6}{3} = \mathbf{2}$

Q16. 4 pts

Line through $(-2, 4)$ and $(2, -4)$.



$$m = \frac{-4-4}{2-(-2)} = \frac{-8}{4} = \mathbf{-2}$$

Q17. 6 pts

Line through $(-1, 5)$ and $(3, -3)$.

$$m = \frac{-3-5}{3-(-1)} = \frac{-8}{4} = \mathbf{-2}$$

Q18. 3 pts

Slope of $y = -4x + 7$?

Coefficient of x: $m = \mathbf{-4}$

Q19. 3 pts

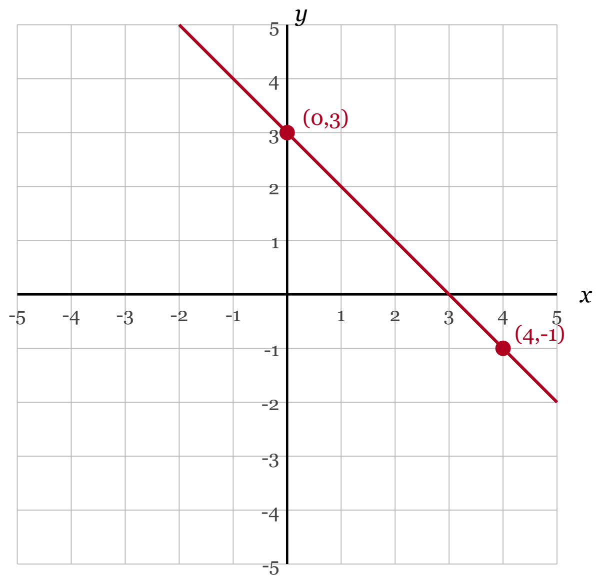
y-intercept of $y = 3x - 5$?

Constant term: $b = \mathbf{-5}$

Section 5 — Graphing Lines 18 pts

Q20. 6 pts

Line through $(0, 3)$ and $(4, -1)$.



(a) $m = \dfrac{-1-3}{4-0} = \dfrac{-4}{4} = \mathbf{-1}$

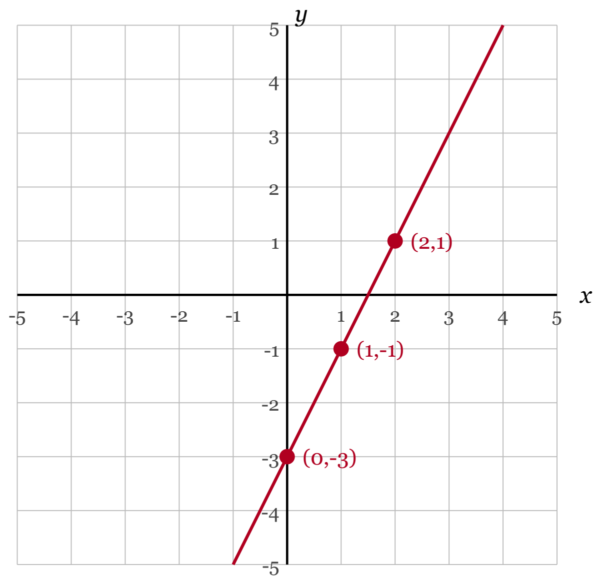
(b) y-intercept = **3**

(c) $y = -x + 3$

Q21.

6 pts

Graph $y = 2x - 3$.

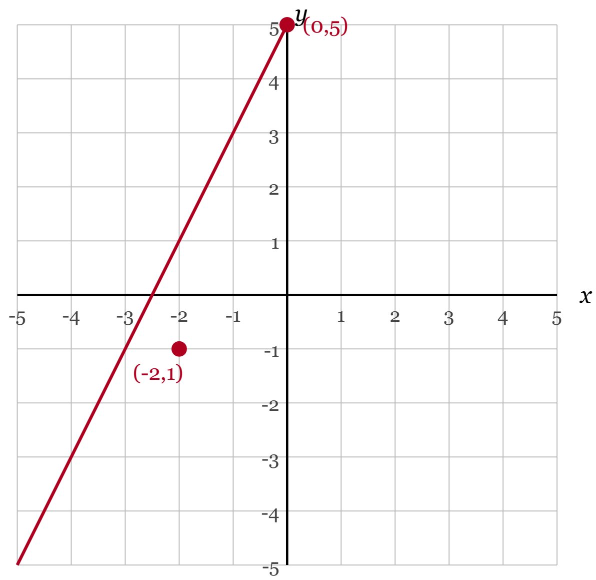


y-intercept: **(0, -3)**. Slope $2 = \frac{2}{1}$: from y-int, go up 2 right 1 to **(1, -1)**, then again to **(2, 1)**. Line continues both directions.

Q22.

6 pts

Table: (-2,1), (0,5), (2,9), (4,13).



Slope: $m = \dfrac{5-1}{0-(-2)} = \dfrac{4}{2} = 2$. y-intercept: 5. Equation: $y = 2x + 5$.

Note: Points $(2, 9)$ and $(4, 13)$ fall above the visible grid (y > 5). Full credit if Sienna draws the line in the correct direction with correct slope, even if she can't plot those two points on the printed grid.

Grading Summary

Section	**Topic**	**Max**	**Earned**	**%**
S1	Evaluating Expressions & Function Notation	14		
S2	Solving Equations	30		
S3	Proportional Relationships	18		
S4	Identifying Slope	20		
S5	Graphing Lines	18		
	Total	**100**		

Placement thresholds:

- **85–100:** Ready for Algebra 8
- **70–84:** Likely ready — review gap areas first
- **55–69:** Borderline — pre-algebra reinforcement recommended
- **Below 55:** Not yet — continue 7th-grade math fundamentals

Concern flags (regardless of total score):

- S2 below 21/30 → equation-solving gap; focus on multi-step, variables-on-both-sides, and solution-type recognition.
- Q10 *and* Q11 both wrong → doesn't yet recognize no-solution / infinitely-many-solutions cases. Spend a week on this before Algebra 8.
- S5 below 13/18 → graphing gap; pre-algebra reinforcement on plotting and slope-intercept form recommended before Algebra 8.